

Tutorial: bibliographic search for writing a survey paper

PR422/PR423 Introduction to research

According to [ACM Computing Surveys](#), “a survey article assumes a general knowledge of the area; it emphasizes the classification of the existing literature, developing a perspective on the area, and evaluating trends.”

I. Objectives

The main objective of this tutorial (TD) is to explore three platforms commonly used for creating a scientific survey paper : [google scholar](#) [Web of science](#) [IEEE](#), [zotero](#) and [overleaf](#).

II. Google scholar

Google Scholar (<https://scholar.google.com/>) is introduced in Wikipedia as “a freely accessible web search engine that indexes the full text or metadata of scholarly literature across an array of publishing formats and disciplines. Released in beta in November 2004, the Google Scholar index includes peer-reviewed online academic journals and books, conference papers, theses and dissertations, preprints, abstracts, technical reports, and other scholarly literature, including court opinions and patents.” [1]

Use the following link to have access to pdf:

<https://login.sid2nomade-1.grenet.fr/login?url=https://scholar.google.com%2f>

Depending on the machine, the link to google scholar may be blocked because of the proxy. You can use IEEE and Web of Science instead :

<https://ieeexplore-ieee-org.sid2nomade-1.grenet.fr>

<https://www-webofscience-com.sid2nomade-1.grenet.fr/wos/woscc/smart-search>

In France, a national platform called [HAL](#) Open Science is also available: <https://hal.science/> [2]

Google Scholar and HAL are examples of platforms that allow researcher to search for peer-reviewed scientific papers.

Accessing scientific resources can also be offered locally by universities which register to scientific publications. This is the case in Université Grenoble Alpes where the platform called [SICD](#) (Service inter-établissements de coopération documentaire) offers an online library: Beluga (Bibliothèque En Ligne de l'UGA : https://beluga.univ-grenoble-alpes.fr/discovery/search?vid=33UGRENOBLE_INST:UGrenoble&lang=fr)

The preceding services are generally accessible through subscriptions.

Other free platforms like <https://arxiv.org/> also exist.

All these platforms can help finding valuable resources for starting a research journey by doing a survey study for example.

Assignment 1 (30mn). Use the Google Scholar, HAL and arXiv platforms to collect some papers related to one of the following topic of interest:

- Integration of sensors into low-power RFID tags
- Data for multi-agent systems and digital twins
- Exploiting the processor pipeline in compiler optimization
- How can safety and security be jointly ensured in embedded systems

Use different combinations of potential keywords and observe the results.

Each student is expected to collect 2 different papers published after 2020, and students of the same group must collect different papers.

III. Zotero

Zotero (<https://www.zotero.org/>) is a reference management platform.

Assignment 2 (15mn) Visit the Zotero related website and discover how to use it. Focus on how to create shared libraries. Use for example the following tutorial: https://www.zotero.org/support/quick_start_guide

Assignment 3 (15mn). Once you have collected references for your research topic, introduce them to your Zotero shared library. Members of each group are required to introduce their 2 references each and to share them with their group.

IV. Overleaf

You will now write a shared summary related to the collected papers. To do this, a shared editing platform like overleaf can be used.

Assignment 4 (5mn). Create an account on <https://plmlatex.math.cnrs.fr/login>

Assignment 5 (5mn). Create a shared project.

Assignment 6 (5mn). Import the IEEE format from: <https://fr.overleaf.com/latex/templates/ieee-conference-template/grfzhnncsfqn>

Assignment 7 (5mn). Create a reference section in your paper, and verify that citations in the paper are correctly numbered and linked to the reference section.

Assignment 8 (to the end). Verify that you can modify, compile and share your document among the team members

V. **How to write a survey paper (optional)**

In the remaining time, you will sketch the outline of a literature review paper based on your preceding search. To this end, both scientific reading and writing skills are used.

S. Keshav [3] proposes the following method for reading scientific literature papers in order to write a survey. You will apply this method to reading your papers. Information below are directly extracted from [3]. During this tutorial, you will apply this reading and writing method.

Assignment 9 (optional):

- 1) Do one quick pass on each paper, then focus on their “related work” sections. You will find a summary of the recent work.
- 2) Find shared citations and repeated author names in the bibliography. These should be the key papers and researchers in that area.
- 3) Download these key papers.
- 4) Check the website of the corresponding authors to find out their recent publications.
- 5) Upload these recent publications.
- 6) Use the papers found in steps 3) and 5) in addition to the papers of your first search to identify key sections of a potential survey paper.

Reference:

[1] https://en.wikipedia.org/wiki/Google_Scholar

[2] [https://fr.wikipedia.org/wiki/HAL_\(archive_ouverte\)](https://fr.wikipedia.org/wiki/HAL_(archive_ouverte))

[3] S. Keshav. “How to Read a Paper”, available at <https://web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf>